

Past Problems

for Korean Undergraduate Mathematics Competition

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This document collects past problems from the Korean Undergraduate Mathematics Competition. General mathematical notation and writing follow *(MATH000) Mathematical Language*.

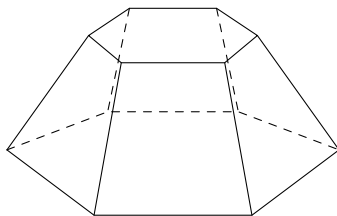
2004.2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. Prove the following.

- (a) If f is continuous at 0, then f is continuous at every $x \in \mathbb{R}$.
- (b) If f is continuous on $\mathbb{R}$ and $f(1) = c$, then $f(x) = cx$ for every $x \in \mathbb{R}$.

2004.2.2. A frustum of a pyramid has parallel upper and lower bases and uniform density. Find the ratio of the distance from its center of mass to the lower base to the distance from its center of mass to the upper base. The similarity ratio of the upper base to the lower base is r , where $r < 1$.



2004.2.3. A convex quadrilateral $ABCD$ has side lengths a, b, c, d , in this order starting from AB . Show that if its area is maximal among all such quadrilaterals, then $ABCD$ is cyclic.

2004.2.4. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$. Find the orthogonal complement of

$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.2.5. Let $f: [a, c] \rightarrow (0, \infty)$ be continuous on $[a, c]$ and differentiable on (a, c) . Let

$$P = (x_0, f(x_0))$$

be a point on the graph of f . Let α and γ be the angles made with the x -axis by the lines joining P to $(a, 0)$ and $(c, 0)$, respectively, and let β be the angle made with the x -axis by the tangent line to the graph at P . Show that there exists $x_0 \in (a, c)$ such that

$$\tan \alpha + \tan \beta + \tan \gamma = 0.$$

If the tangent line does not meet the x -axis, take $\beta = 0$.

2004.2.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

$$\sum_{n=1}^{\infty} a_n^2$$

converges. If

$$\sum_{n=1}^{\infty} f(a_n)$$

converges, prove that

$$\sum_{n=1}^{\infty} |f(a_n)|$$

also converges.

2004.2.7. Suppose a plane curve $(x(t), y(t))$ of period T satisfies

$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy, \quad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.2.8. Evaluate

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx.$$

2004.2.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

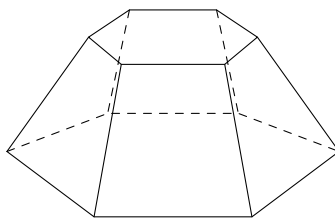
2004.1.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

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$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.1.5. Let A be the least positive integer of the form

$$33^k - 7^\ell, \quad k, \ell \in \mathbb{N},$$

and let B be the least positive integer of the form

$$7^m - 33^n, \quad m, n \in \mathbb{N}.$$

Find $A + B$.

2004.1.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

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converges. If

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where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.1.8. Let $\mathbf{v} \in \mathbb{R}^n$ be a unit vector, and let $\mathbf{P}_{\mathbf{v}}$ denote the orthogonal projection onto the line $\text{span}(\mathbf{v})$, so that

$$\mathbf{P}_{\mathbf{v}}\mathbf{x} = (\mathbf{x} \cdot \mathbf{v})\mathbf{v}$$

for every $\mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{A}$ be an $n \times n$ positive definite symmetric matrix. Find the maximum value of λ such that

$$\mathbf{A} - \lambda \mathbf{P}_{\mathbf{v}} \succeq 0.$$

Here $\mathbf{B} \succ 0$ means that every eigenvalue of the symmetric matrix $\mathbf{B}$ is positive, and $\mathbf{B} \succeq 0$ means that every eigenvalue is nonnegative.

2004.1.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

2003.2.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2003.2.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC, BCD, CDA , and DAB , respectively. Prove that $PQRS$ is a rectangle.

2003.2.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.2.4. Let (x_n) be a sequence satisfying

$$0 \leq x_{n+1} \leq x_n$$

for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} x_n$$

converges. Determine

$$\lim_{n \rightarrow \infty} nx_n.$$

2003.2.5. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ satisfy

$$x_1 + x_2 + \dots + x_n = 0.$$

Prove that

$$\max_{1 \leq j \leq n} |x_j| \leq \left(\frac{n-1}{n} \right)^{1/2} \|\mathbf{x}\|_2,$$

where

$$\|\mathbf{x}\|_2 = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}.$$

2003.2.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, \ k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.2.7. Prove that the real number $\sin 1$ is irrational.

2003.2.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3}\right)^n (n+1)$$

for every $x \in A$.

2003.2.9. Let f be the 2π -periodic function defined by

$$f(x) = \begin{cases} \pi + x, & -\pi < x \leq 0, \\ \pi - x, & 0 < x \leq \pi. \end{cases}$$

Using the Fourier series of f , prove that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \cdots + \frac{1}{(2n-1)^2} + \cdots.$$

2003.1.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

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2003.1.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.1.4. Let (a_n) be a sequence with $a_n \geq 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n$$

converges. Prove that, for every $p > \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

converges. Also give an example showing that, when $p = \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

may diverge.

2003.1.5. Let $p \geq 1$ be a real number, and let $x_1, x_2, \dots, x_n > 0$ satisfy

$$x_1 + x_2 + \dots + x_n = 1.$$

Prove that

$$\sum_{j=1}^n \left(x_j + \frac{1}{x_j} \right)^p \geq \frac{(1+n^2)^p}{n^{p-1}}.$$

2003.1.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.1.7. Prove that the real number $\sin 1$ is irrational.

2003.1.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3}\right)^n (n+1)$$

for every $x \in A$.

2003.1.9. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix whose entries are all 0 or 1. Suppose that, for some $t \in \mathbb{N}$,

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} t+1 & 1 & \cdots & 1 & 1 \\ 1 & t+1 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & t+1 & 1 \\ 1 & 1 & \cdots & 1 & t+1 \end{pmatrix} = t\mathbf{I}_n + \mathbf{J},$$

where $\mathbf{A}^\top$ denotes the transpose of $\mathbf{A}$, $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\mathbf{J}$ is the $n \times n$ all-ones matrix. Express $\det(\mathbf{A})$ as a function of t .

2002.0.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Suppose the function $x \mapsto xf(x)$ is differentiable for $x \neq 0$ and

$$\frac{d}{dx}(xf(x)) = e^x.$$

Find f .

2002.0.2. Let (a_n) be an arithmetic sequence with first term a and common difference d , and define the 2002×2002 matrix

$$\mathbf{A} := \begin{pmatrix} a_1 & a_2 & a_3 & a_4 & \cdots & a_{2002} \\ a_2 & a_1 & a_2 & a_3 & \cdots & a_{2001} \\ a_3 & a_2 & a_1 & a_2 & \cdots & a_{2000} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2002} & a_{2001} & a_{2000} & a_{1999} & \cdots & a_1 \end{pmatrix}.$$

Express $\det(\mathbf{A})$ in terms of a and d .

2002.0.3. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be unit vectors in $\mathbb{R}^n$. Prove that

$$|\langle \mathbf{a}, \mathbf{c} \rangle \langle \mathbf{b}, \mathbf{d} \rangle - \langle \mathbf{a}, \mathbf{d} \rangle \langle \mathbf{b}, \mathbf{c} \rangle| \leq 1,$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^n$.

2002.0.4. Let $n \in \mathbb{N}$, and let $a_1, \dots, a_n, x_1, \dots, x_n$ be positive real numbers satisfying

- (i) $a_1 \leq a_2 \leq \cdots \leq a_n$;
- (ii) $x_k \geq x_n$ for $k = 1, 2, \dots, n-1$.

Prove that

$$(a_1 + x_1 a_2)(a_2 + x_2 a_3) \cdots (a_{n-1} + x_{n-1} a_n)(a_n + x_n a_1) \geq a_1 a_2 \cdots a_n (x_1 + 1)(x_2 + 1) \cdots (x_n + 1).$$

2002.0.5. For $\alpha \in \mathbb{C}$, define the $n \times n$ matrix $\mathbf{A}_n = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & i = j, \\ \alpha, & i - j = -1, \\ \bar{\alpha}, & i - j = 1, \\ 0, & |i - j| \geq 2. \end{cases}$$

Prove that $\mathbf{A}_n$ is positive semidefinite for every $n = 1, 2, \dots$ if and only if

$$|\alpha| \leq \frac{1}{2}.$$

Here $\bar{\alpha}$ denotes the complex conjugate of α .

2002.0.6. Solve the system of differential equations

$$x''(t) + 2y'(t) - x(t) = 0, \quad y''(t) - 2x'(t) - y(t) = 0$$

subject to

$$x(0) = y(0) = 0, \quad x'(0) = a > 0, \quad y'(0) = 0.$$

2002.2.1. For $(x, y) \neq (0, 0)$, define

$$f(x, y) = \frac{x^2 - xy + y^2}{2x^2 + y^2}.$$

Find the minimum value of f .

2002.2.2. Let $f, g, h: [-1, 1] \rightarrow \mathbb{R}$ be continuous. Suppose that f is odd, g is even, and

$$\int_{-1}^1 f(x)^2 \, dx = \int_{-1}^1 g(x)^2 \, dx = 1.$$

Prove that

$$\left(\int_{-1}^1 f(x)h(x) \, dx \right)^2 + \left(\int_{-1}^1 g(x)h(x) \, dx \right)^2 \leq \int_{-1}^1 h(x)^2 \, dx.$$

2002.2.3. Let $a > b > 0$. Evaluate

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} \, dx.$$

2002.1.1. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative, with $f(a) = 0$. Suppose that

$$\left| \frac{f'(x)}{2003} \right| \leq \left| \frac{f(x)}{2002} \right| \quad (x \in [a, b]).$$

Prove that $f(x) \equiv 0$ on $[a, b]$.

2002.1.2. Let $n \in \mathbb{N}$ with $n > 1$. Prove that

$$\frac{((n-1)!)^{1/(n-1)}}{(n!)^{1/n}} < \frac{(n!)^{1/n}}{((n+1)!)^{1/(n+1)}}.$$

It is given that

$$4 > e^{4-2\sqrt{2}}.$$

2002.1.3. Let

$$\alpha = 0.0110101000101000101\dots,$$

where the n th digit after the decimal point is 1 if n is prime and 0 otherwise. Prove that α is irrational.

2001.2.1. Let $0 < a, b < 1$ satisfy

$$\int_0^a \frac{1}{\sqrt{1-x^2}} dx = \int_b^1 \frac{1}{\sqrt{1-x^2}} dx.$$

Find $a^2 + b^2$.

2001.2.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

2001.2.3. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2y = 0, \quad x > 0.$$

Hint: $y = e^{x^2}$ is a solution of the differential equation.

2001.2.4. Let $f: [0, \infty) \rightarrow [0, \infty)$ satisfy

$$2f(x)f(1) \leq f(x)f(1) + 1 \leq 2f(2x)$$

for every $x \geq 0$. Prove that f is constant.

2001.2.5. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative. Evaluate

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(x) \left(|\sin(\lambda x)| - \frac{2}{\pi} \right) dx.$$

2001.2.6. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C , respectively. Find the maximum value of

$$a \cos A + b \cos B + c \cos C.$$

2001.2.7. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2} \right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.2.8. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ be a unit vector. Prove that

$$\sum_{i=1}^n \|\mathbf{x} - \mathbf{v}_i\|^2 \geq 2n - 2\sqrt{n}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

2001.2.9. Let $\lambda \geq 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2001.1.1. Show that there are infinitely many triples $(a, b, c) \in \mathbb{N}^3$ satisfying

$$a \cos \left(\arctan \frac{b}{a} \right) + b \sin \left(\arctan \frac{b}{a} \right) = c.$$

2001.1.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

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for every $x \geq 0$. Prove that f is constant.

2001.1.5. Evaluate

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} \frac{\sin x}{n^2 \left(x - \frac{\pi}{4}\right)^2 + 1} dx.$$

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$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^2$ satisfy

$$\|\mathbf{v}_i\| \leq 1 \quad (i = 1, 2, \dots, n).$$

Prove that there always exist signs

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{1, -1\}$$

such that

$$\|\varepsilon_1 \mathbf{v}_1 + \varepsilon_2 \mathbf{v}_2 + \dots + \varepsilon_n \mathbf{v}_n\| \leq \sqrt{2}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

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$$f(x) = (x - \lambda f(x))^p.$$

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